2015 Fifth International Conference on Instrumentation and Measurement, Computer, Communication and Control

ATS Software Framework Design Pattern and Application

Yuan Mei
School of Automation Science and Electrical Engineering
Beijing University of Aeronautics and Astronautics
Beijing, China
yuanm@buaa.edu.cn

Fan Lingjie
School of Automation Science and Electrical Engineering
Beijing University of Aeronautics and Astronautics
Beijing, China
fanling5458@126.com

***Abstract*—Framework is a way of reusing the design of whole system or part of it, which is considered as the most effective way now in software engineering. The trend of automatic test system (ATS) software is becoming more and more complex. In response to this trend and to improve the efficiency of its software development, combined with the technology related to software engineering and automatic test technology, this paper presents a pattern of developing software framework for the ATS, which is independent of the hardware system. By the instance of building the CAN test system software framework, it has proved that this pattern works well. The proposed method can help developers to build domain software framework quickly, to cut down the development cycle greatly, and to reduce the development cost obviously.**

Keywords—software framework; automatic test system; software reuse; CAN test system

I. Introduction

Automatic Test System (ATS) refers to a system that can automatically complete certain or complex measure tasks under some program control instructions. It includes measurement instruments and other equipment, but the computer is the most important part in these [1]. ATS usually have two parts, the hardware and software. The hardware constitutes the basis of ATS, and the software is the soul of it. With the rapid development of science and technology, the ATS has become more and more complex and the scale of integration has increased larger and larger. What's more, standard interface bus makes ATS hardware platform to be built and restructured more conveniently. The construction process of the system is accelerated. Hardware compatibility of the system has been improved [2].In this case, the status of the software has never got so much attention before. Under the influence of virtual instruments, there has been the idea of "hardware to be replaced by software". A hardware platform can be set up quickly by a simple combination of standard hardware. The specific functions are defined and implemented by software. As the core and key parts of the test system, ATS software is the bridge and central of the unit under test (UUT) and test resource. It has a direct impact on the overall performance of the system [3].

With the improvement of the situation in system develop, ATS software has become increasingly large and complex, therefore, traditional software developing methods, such as process-oriented, are not applicable any more. It is difficult to complete the ATS software in an economic way by using traditional methods. Further, it's a hard task to maintain, extend and reuse the software. There is always a long repeat of work.

In the expertise area of software, there is an idea of software framework in order to solve the problems above. This idea has been proved that it works well. Software framework is soft-reuse technique for certain field. However in the ATS software, there are particular problems to apply these techniques to ATS software due to the limitations of the developers. Not to mention to develop the special software framework design pattern for automatic test system according to its characteristics [4].

This paper introduces a pattern to develop ATS software framework which is on the basis of the existing ATS software development experience. And it is combined with relevant theories of software framework and the methods of software engineering. By applying it to a multi-channel CAN test system development, this paper describes the design principles, methods and technology to follow in each stage and what should be paid most attention. And it presents the concrete form to achieve the goal and been applied to each stage.

II. Common Method for ATS Software Framework Development

Software framework is soft-reuse technique for certain field. It is a reusability design of an application software for reusing part or all of this software. It is usually presented by an interaction way of a set of abstract classes and instances [5].

In order to improve the efficiency of software development, software framework is an effective solution. It is a combination of design reuse and code reuse. Unlike the class libraries only providing simple code reuse, software framework can provide the reuse of the whole process of building system, from beginning to the end, including design, components, models, code organization, code design, code implementation, system combination and so on. It can realize the accumulation of experience and reduce the cost, lessen the duplication of work [6].

978-1-4673-7723-2/15 $31.00 © 2015 IEEE
DOI 10.1109/IMCCC.2015.37

In this paper, it has provided a common pattern to develop ATS software framework for automatic test system. First, the system should be analyzed in order to get the common function module models. The model and its implementation are described using UML which support the reuse of the model design. Then, by developing a class library or components to implement the modules. During the actual development, some common problems should be summarized to the Design Patterns. The developed common components are embedded into the development platform such as Microsoft Visual Studio 2010. In this way, the components are really reused. At last, the framework for the special area could be built. It should establish the hierarchical structure and module framework to increase the maintainability and extensibility with reference to the idea of MVC (Model View Controller). After the framework is built, ATS software should be developed under the framework so that the framework is tested and improved. In addition, since it is for ATS field, it is necessary to refer to the standard of ABBET (A Broad-Based Environment for Test) when designing the components and interface.

The develop pattern is shown in Figure 1:

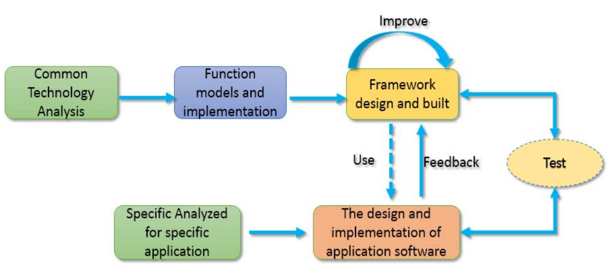


Fig. 1. A Common pattern to develop ATS software framework.

Function models and implementation is a very important step in this pattern. UML (Unified Modeling Language) is the unified modeling language or standard modeling language. It is a graphical language that supports model and software system development describe. UML can provide modeling and visualization support for all stages of software development, including requirements analysis to the specifications and from the structure to configuration [7].

Component technology is one of the most important software reuse technologies. Process-oriented programming could reuse functions. Object-oriented programming could reuse classes. Generic programming could reuse algorithm source code. Different from these, component could reuse the whole specific function programming module [8]. Its architecture is defined by the component model and how to use this architecture and interact with other subjects.

Design Pattern describe the core issues and the solutions of the problems which encountered repeatedly. So that we can reuse the pattern to avoid repeating work [8]. In general, a pattern has four basic elements: pattern name, problem, solution and consequences. Depending on the character of the pattern, it is divided into three types: Creational, Structural and Behavioral.

In this paper, when designing the class libraries or the interfaces, the solutions of common problems should be summed up to relevant design patterns. For example, if the system has only one device when it is running, we can use the design pattern: Singleton. It means there is only one instance in the system which makes sure the device can be controlled only by this instance. By this way, it can avoid competition and confusion.

III. Software Framework for CAN Test System

In this paper, the software framework of CAN test system is developed as a living example. Using the previous ideas and technologies, the most important design principles should be followed and methods or techniques used in every stage of the process are introduced.

The platform used is Visual Studio 2010. The main develop language is C#, and some C++/C.

A. Common technology analysis

Although the framework is for CAN test system, it strives to have a certain versatility in the test system field by combining the development experience of many other test systems.

The common technology analysis includes both the project and the field, so that it could establish common models. As shown in figure 2, there are some common models for common test system.

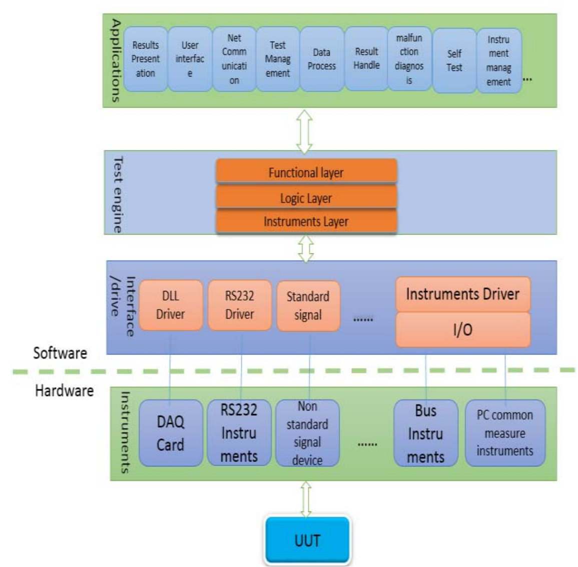


Fig. 2. Common models for ATS

The most important principle in this part is to pack the change part. So that, this will increase the independence between the modules and enhance reusability.

B. Function models and implementation

(1) Function models

UML is used to describe the models for function models. It accords with standard and easy to modify and expand the system. UML can describe the design of the module and the relationship between modules. As shown in figure 3, it describes a data processing module of CAN test system and the relationship between classes.

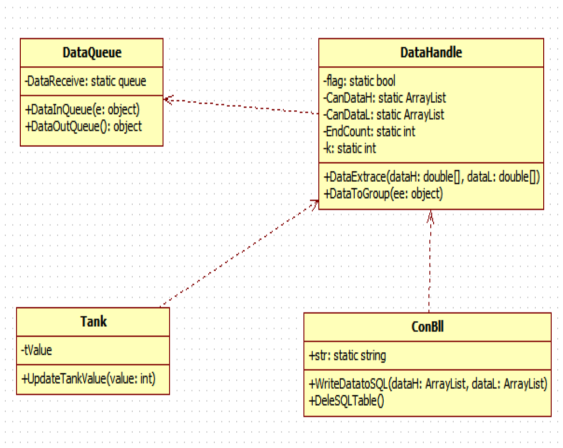


Fig. 3. function models described used UML

Loose coupling should be paid most attention during module design. Modifiability and scalability are also important. ABBET should be referred during function module design and extract, hierarchical structure and interface, etc.

(2) Implementation

Function module implementation is mainly through class libraries and user-defined components. According to the ideas mentioned above, the CAN test system are divided into these components: communication components, interface components, user management module, waveform display module, parameter settings and other components, and drive libraries, data processing libraries, etc. Common problems should be attributed to the appropriate design patterns so that the experience of solving problem can be accumulated and reused.

- Class libraries development

When developing the class library, hierarchies and one-class-one-function should be cared. It means that a clear hierarchy and each library should be responsible for single function or be single level related. And libraries should not be mixed in order to provide good modified ability and scalability which reduces the coupling between them. In general, 3-tier architecture is as reference which includes Presentation layer, Business Logic layer and Data access layer [9]. For example, when using general hardware, the classification could refer to FIG.4, which also marked the role of each level. Similar to other libraries such as databases related libraries, the difference is that the object is a database.

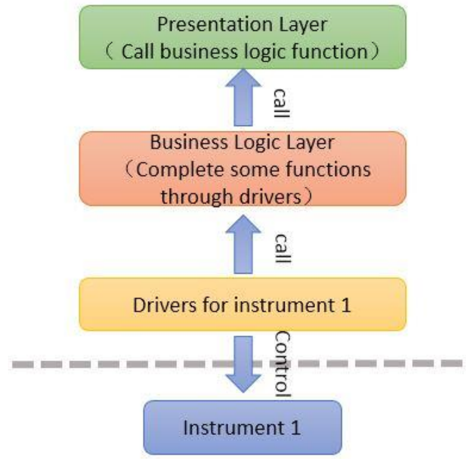


Fig. 4. 3-tier architecture example for an instrument

- Components development

Class libraries can describe models and some their features, but the code is abstract. According to the models built by previous analysis, models could be developed into independent components. They provide simple and visual reuse forms, not reuse just in concept any more, but presented in common development platform. As the platform Visual Studio 2010, components are embedded in the platform as user controller. Such as the configuration component, if it is embedded in Visual Studio 2010 as a controller, in order to reduce repeat work, we can reuse it by dragging it into the form when developing similar systems. During this process, related Design Pattern should be summarized in order to reuse the solutions of similar problems and provide good interfaces of the component.

In Visual Studio 2010, the step to build custom control library is as follows: “File – new – Project – Windows Custom Control Library”. And then we could implement the logic function in the control library, the component in the development could be reused as a controller.

In this project, these components are often used, configuration, wave display, data display and so on. As show in FIG5, it is a wave display controller, which receives data and displays by wave. It can be embedded in Visual Studio control library. Then it can be reused by a way of visual and dragging. The control library can be developed by yourself and can also use external resources, such as NI Measurement Studio.

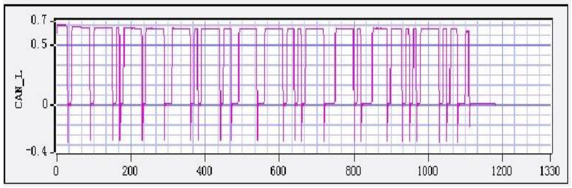


Fig. 5. an example of self-develop controller

When completing the class library and component, the coding should be focused on the interface not the realization, following common design patterns.

C. *Framework design and built*

Framework design has a main reference, the idea of MVC (Model View Controller). As the FIG6 show, the framework and modules are organized by Model - View – Controller in order to reduce the coupling between them [10]. For the CAN test system, the View mainly includes wave display, data show, analysis result and so on. The models are these models in 3.1. The controller's work is mainly to determine the model state, such as the data updates of CAN protocol modules. The View mainly presents the state and provides the user interface.

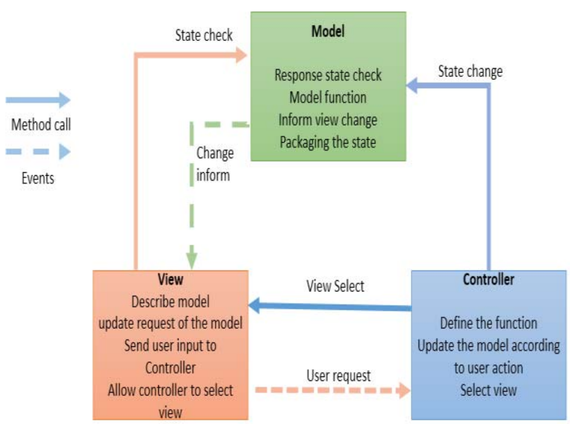


Fig. 6. MVC components and their relationship

After the analysis introduced in 3.1, framework integration is to build the relationship between the models and the organization of layers to form the required framework. The actual presentation is: the analysis and design are described by UML, the code is reused by class library and the components embedded in development platform. Thus, the framework is built.

When developing specific application under the framework, such as the CAN test system in this project, the application could be completed quickly through the invocation and the combination of the models in the framework. At the same time the application could also give feedback to the framework so that it is modified, accumulated and expanded, then the framework could also be used in other applications of this field. This framework is presented in abstract as in FIG 7.

In this figure, the communication and driver component library are the drivers of the hardware used frequently and the common communication way used often, such as COM. User interface component library not only includes the buttons and labels which the platform bring by own but also includes the usual controller used often in our own project, such as the wave display component. Data-related component library mainly has the class libraries and components to access to the database or the data file often used in project. General measurement-related component library has the components for common requirement we used frequently in this specific field. And it is expanded as the experience is growing. Other components libraries are for these which are difficult to be classified but reuse is important.

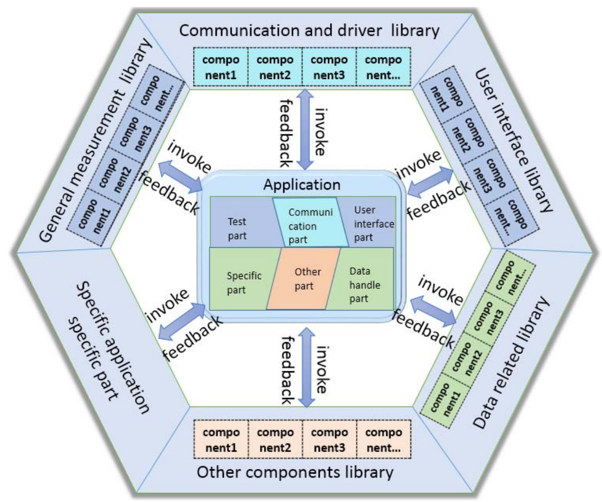


Fig. 7. Abstract Framework

The overall design of the framework and its design should pay attention to its commonality, extensibility, and modifiability.

D. *Framework Test*

The framework could not be perfect at the beginning, need constantly modify and improve. Thus it needs to be tested frequently. There are mainly two ways, one is active, extends and seeks defects of the framework forwardly. And the other is passive, that is to develop specific applications under the framework. Modify and improve the framework by getting the feedback in the development process. Thus, when developing applications under the framework, in addition to the specific analysis, feedback to the framework is also very important. As in the figure 7, every library has feedback in order to improve the framework.

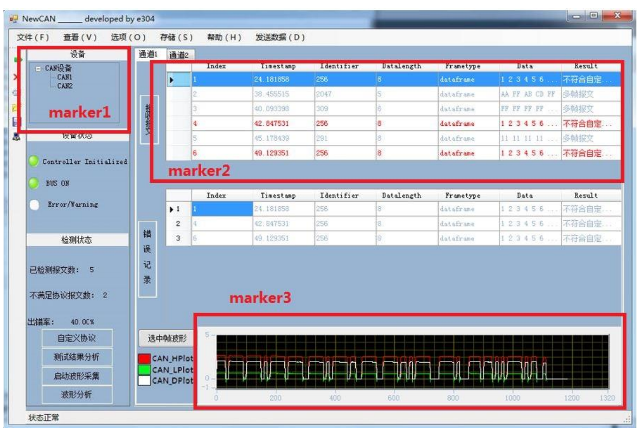
In this stage, the principle is that combination should be used as much as possible instead of inheritance when testing or improving the framework by developing specific applications. This can reduce hierarchy and complexity.

IV. MULTI-CHANNEL CAN TEST SYSTEM SOFTWARE IMPLEMENTATION

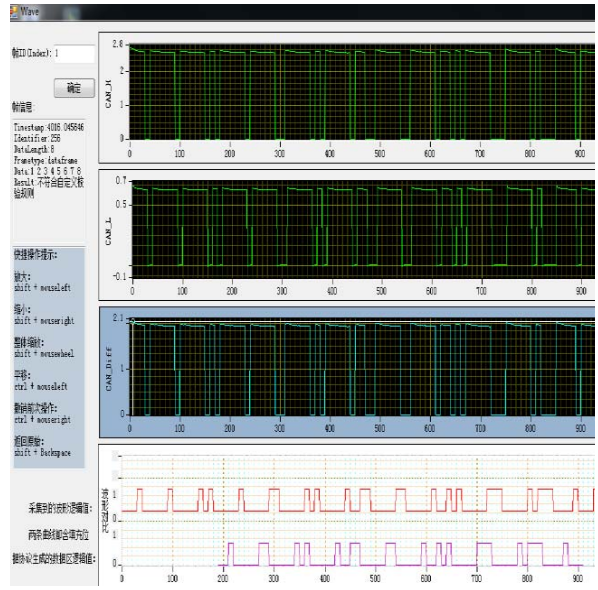
To develop multi-channel CAN test system under the framework completed, the first step is to make specific analysis of the system. This multi-channel CAN test system has 4 channel and its main function is to test the communication message, get and store their physical waveform, play back the data and wave, carry out protocol analysis for each channel.

This system has many channels, so many similar forms could be reused, that means there are a lot of reuse situations. The models which could be reused under the framework are data manage controller, communication controller, data storage controller and so on. These models could be reused after being simply modified. The models which should be added are wave get model, protocol analysis model and so on. And then add the new models to the framework. These new models could be used many times but just need to be developed once, it could be beneficial for this project itself. The related class libraries could also be reused by giving different parameters and give out different results of different channel.

Through the combination and improvement of the framework, models could be integrated into the control library of the development tool Visual Studio 2010. Then the software could be completed rapidly and its modifiability and scalability are very good. No more details, the final develop result of the software is as FIG 8.



(a) Main form of the software



(b) the wave display and analysis form

Fig. 8. multi-channel CAN test software test system

In the development, FIG 8(a) marker 1 is that the software reuse driver classes library. Marker 2 is that the software use the display controller in Visual Studio 2010 directly. Marker 3 is the wave display controller we developed in this project and reuse it many times in FIG 8 (b) and could be reused in other similar projects. In this project, the main function of the form in FIG 8 (a) is to get the CAN message, make analysis, bus state and show a simple wave of the selected message. The main function of the form in FIG 8 (b) is the CAN High, CAN Low and CAN Diff wave in detail and to support detailed view by zoom, pan etc. On the bottom of the FIG 8 (b) is the wave comparison, which compare the logic value of the physic wave with the logic value of the protocol data. Attention should be paid to the rule of the CAN message padding bit.

FIG 9 is the workload of the project, comparing plan days with actual work days needed under the framework. In order to be fair, they are counted after the key problems have solved. The plan days are on the basis of the past similar projects with no framework. Although there is some subjectivity, it has told something. We can see from the FIG9, the work under the framework has an distinct advantage. This advantage could be more obvious with the improvement of the framework.

Main Category	Small Category	Workload estimation on the basis of the experience（day）	Actual workload under the framework（day）
Requirements analysis and design	Demand survey	5	5
	Demand analysis	4	4
	Design	5	3
Software system design	Architecture design	4	2
	Outline design	6	4
	Detailed design	20	12
	Database Design	5	2
	Other code implementation	10	7

Fig. 9. Work days of some categories

On the process of the development, it could also expand and modify the components of the framework. Then the future development will get benefits and reuse these components. The components will be more and more, the framework will be richer, the application development under the framework will be quicker.

V. Conclusion

In this paper, it has explored a general pattern to develop ATS software framework and used it in the development of software framework for CAN test system. Compared with other development ways, developing applications under software framework could improve the efficiency and the quality obviously. The develop pattern in this paper for the ATS software framework is convenient and efficient, from the beginning to the end. And it has strong modifiability and scalability. Other similar systems could also use it to develop applications. Different from other ways like object-oriented only, it is more specific and systematic. For example, 3-tier architecture and making the components embedded into the development platform rather than just reusing them in abstract. The framework works well and could be improved and expanded constantly.

The framework design pattern in this paper emphasizes the combination of software engineering technology and the knowledge in ATS field. And pay attention to software design reuse and system scalability. The software framework developed could achieve reuse in every stage from the design to code implementation. Through developing an specific application, it has proved that developing ATS software under software framework could shorten the development cycle and reduce the cost obviously. What's more, the software could get good quality and robust character.

References

[1] Guo RongBin, Zhao XiuCai. The trend of automatic test system[J]. Foreign Electronic Measurement Technology, 2014, 33(6)(In Chinese)

[2] YANG Aibing, SUN Ye, PENG Wei. Research on the software development model of automatic test system[J]. INDUSTRIAL INSTRUMENTATION & AUTOMATION. 2010(6). 16-18。

[3] Liu Q. Design of universal ATS software framework based on signal[J]. Electronic Measurement Technology, 2012.

[4] Wang Y P, Wang D G, Zhong-De W U. Signal-Oriented ATS Test Software Structure and the Model[J]. Journal of Naval Aeronautical & Astronautical University, 2013.

[5] Liu Qi, He Yuzhu. Design of universal ATS software framework based on signal[J]. Electronic Measurement Technology, 2012, (12):46-49.

[6] Sun Xiaojin, Guo Enquan. Design of software architecture for fault diagnosis system based on IEEE1232[J].Journal of Electronic Measurement and Instrumentation, 2014, 28(1):36-42.

[7] Li Chuan-Huang, WANG Wei-Ming, SHI Yin-Yan. Performance Prediction Method for UML Software Architecture and Its Automation[J]. Journal of Software, 2013, (7):1512-1528.

[8] LIU Feng, SUN Yong. Application of Design Patterns and Component Technology to Business Logic Layer[J]. Computer Systems & Applications, 2011, 20(10):154-159.

[9] ZHANG Yuan, ZHANG Zhao, LIU Rui. Modular design of virtual experiment system based on MVC design model[J]. Computer Engineering & Science, 2013, 35(8):125-129.

[10] Guo Shungzhou, Liang Jinlan. Design and Implement ation of Automated Testing Framework[C].Computer Measurement & Control. 2009:224-227.

[11] LIU Jin Ning, MENG Chen, HOU Yan. Research of ATS software model[J]. Measurement Control Technology and Instruments, 2007, 33(9): 81-84.

[12] WANG Ying-qiang, CHEN Sui-yang, WANG Zheng-feng. Design and Implementation of Reusable Software Development Framework Based on . Net[J]. Computer Technology and Development, 2014, (6):122-126.